



جامعة الطائف  
TAIF UNIVERSITY

## TAIF UNIVERSITY

College of Computers and Information Technology  
Department of Computer Engineering



# NETWORKING PROJECT

— Secure Satellite Communication and IoT Environmental Monitoring System —

Course: Computer Networks

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# Introduction



The Secure Satellite Communication and IoT Environmental Monitoring System is a smart prototype that combines network security and embedded systems. The network was designed using Cisco Packet Tracer with VLANs and security controls, while an ESP32 collects data from motion and temperature sensors to monitor the environment and generate alerts. The project demonstrates how networking and embedded technologies can work together to improve system security, monitoring, and reliability.

# Project Description

Our project is a secure satellite communication system integrated with a smart environmental monitoring system. The main goal is to simulate a ground control station that continuously receives data and monitors both the system status and the surrounding environment.

First, we designed the network using Cisco Packet Tracer. It includes a router, switches, and multiple devices, which are divided into VLANs to enhance security and organize communication between devices.

On the Embedded Systems side, we used an ESP32 microcontroller along with several sensors, such as:

1. A PIR sensor for motion detection
2. A DHT11 sensor for measuring temperature and humidity

The ESP32 reads and processes the sensor data to determine the system status. If motion is detected or the temperature rises beyond a certain level, an alert is triggered.

In general, the network is responsible for data transmission and securing communication, while the embedded system handles data collection and environmental monitoring. This integration makes the system smarter and more secure, representing a simplified model of modern control center systems.

# Objectives

1. provide real-time monitoring of the system
2. generate fast alerts in case of any danger or abnormal activity
3. simulate a secure protection and monitoring system for ground control stations



# Network Topology

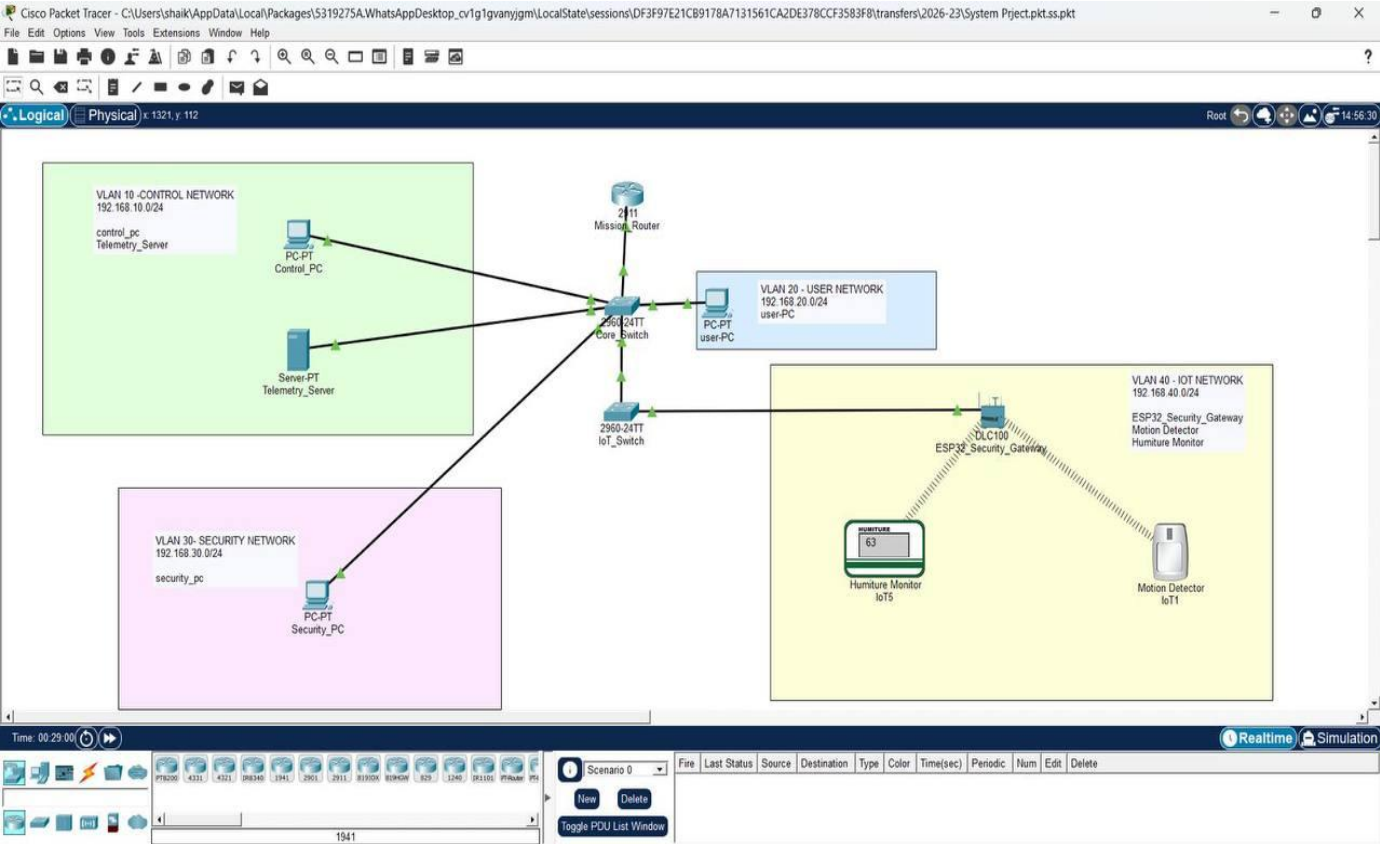


Figure 1 Cisco Packet Tracer Topology

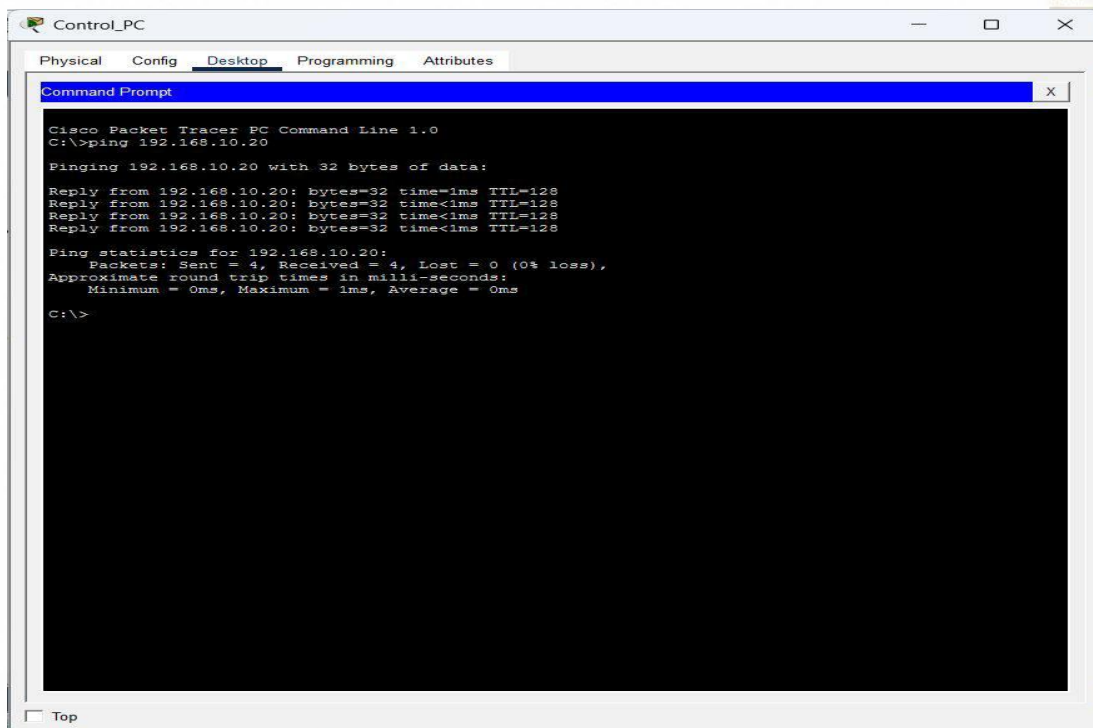


Figure 1 Ping Success

## 1.Successful Ping Communication :

The communication between the Control PC and the Telemetry Server was successfully verified using the Ping command.

Result :

- \* Packets Sent = 4
- \* Packets Received = 4
- \* Packet Loss = 0%

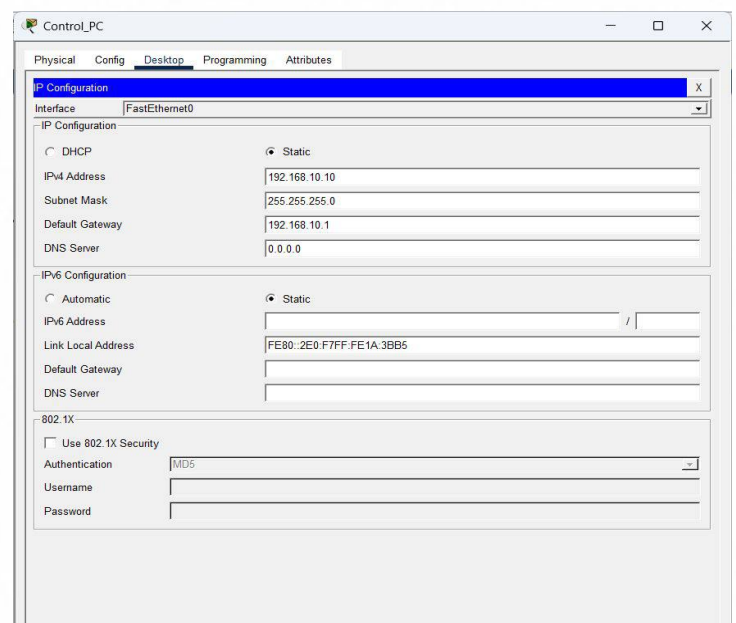


Figure 2 Ping Success

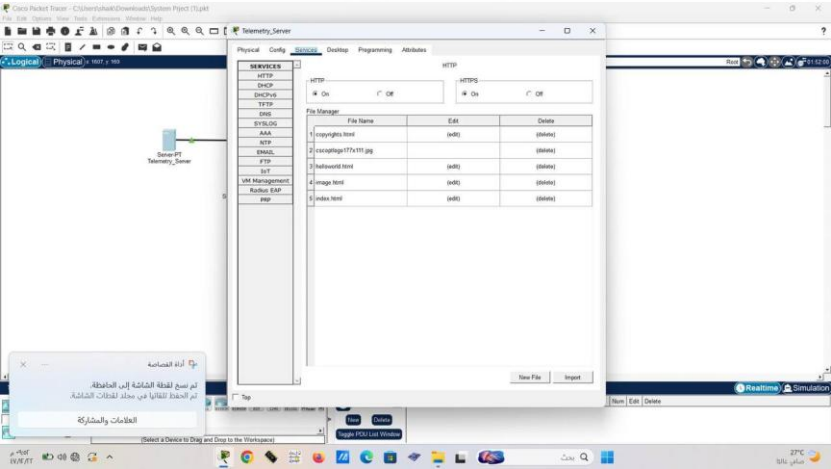


Figure 3 Server Operation

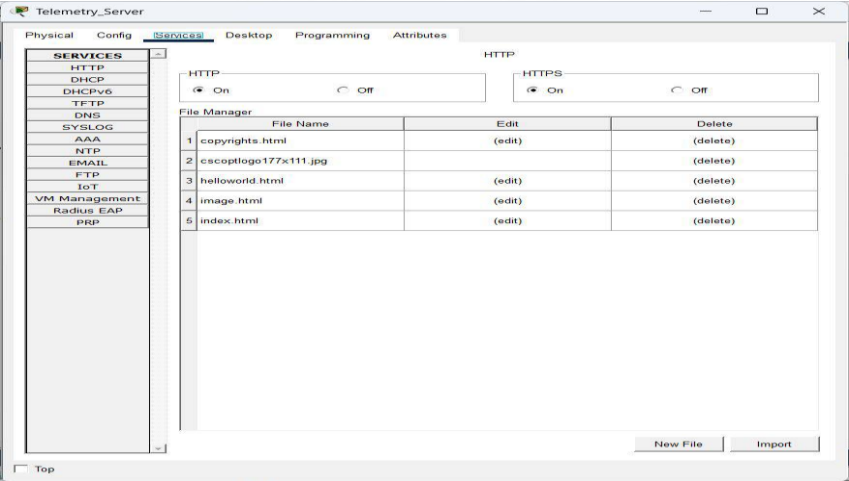


Figure 4 Server Operation

## 2.Server Operation:

### Telemetry Server Operation

The server is operating correctly and its essential services are enabled.

Enabled Services :

- \* HTTP ✓
- \* HTTPS ✓



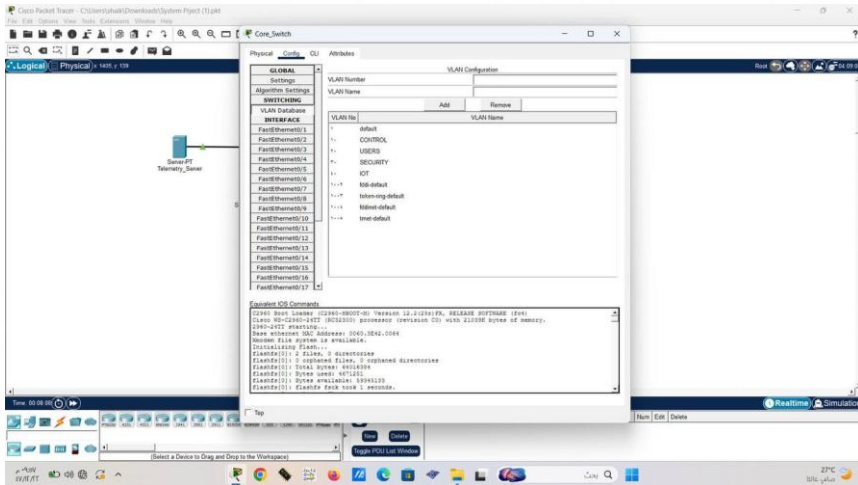


Figure 6 security implementation

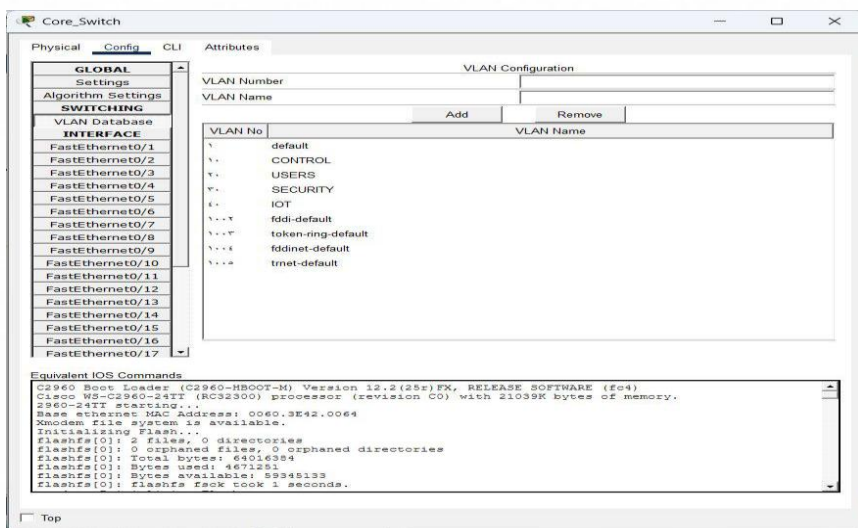


Figure 5 security implementation

### 3.Security Implementation:

Security was implemented through VLAN segmentation to isolate devices and organize network traffic.

Configured VLANs :

- \* VLAN 10 → Control Network
- \* VLAN 20 → User Network
- \* VLAN 30 → Security Network
- \* VLAN 40 → IoT Network

# Project components



Table 1 Hardware components

| No. | Component                       | Quantity |
|-----|---------------------------------|----------|
| 1   | Cisco 2911 Router               | 1        |
| 2   | Cisco 2960-24TT Switch          | 2        |
| 3   | Server-PT (Telemetry Server)    | 1        |
| 4   | Control PC                      | 1        |
| 5   | Security PC                     | 1        |
| 6   | User PC                         | 1        |
| 7   | Home Gateway                    | 1        |
| 8   | Motion Detector (PIR)           | 1        |
| 9   | Humidity and Temperature Sensor | 1        |
| 10  | ESP32 Security Gateway          | 1        |

Table 2 Network Components

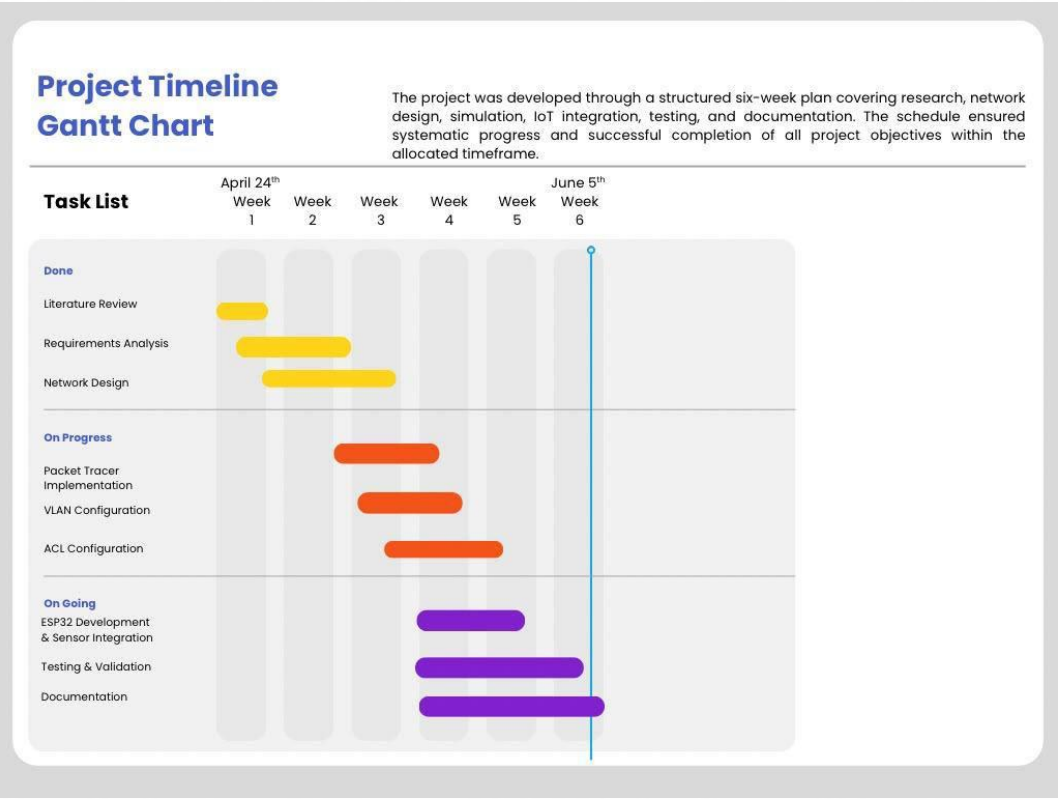
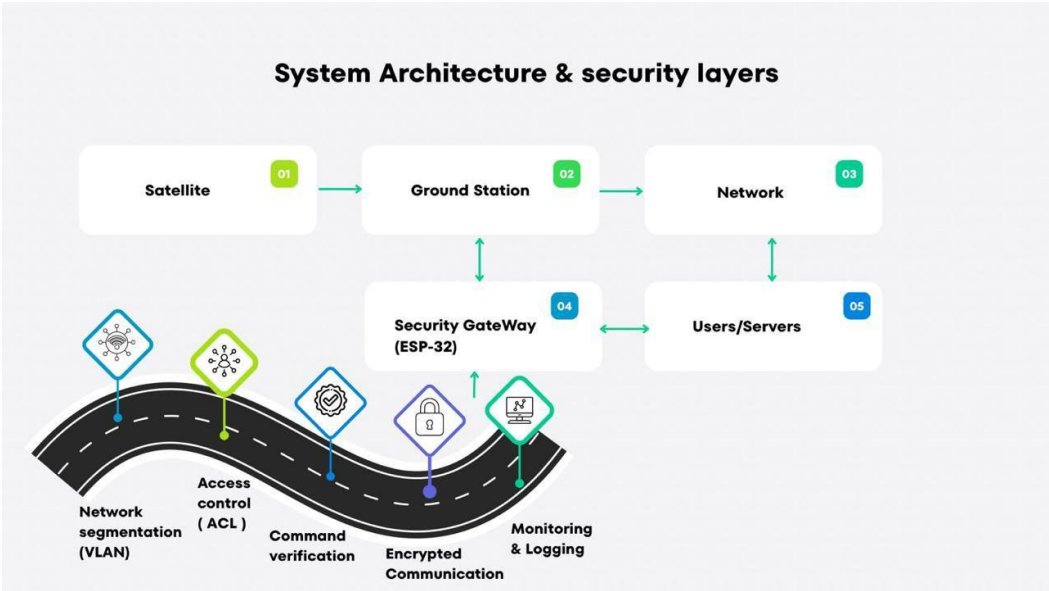
| No. | Component                           | Quantity | Description                                                                |
|-----|-------------------------------------|----------|----------------------------------------------------------------------------|
| 1   | ESP32 Development Board             | 1        | Main microcontroller used for sensor monitoring and wireless communication |
| 2   | PIR Motion Sensor (HC-SR501)        | 1        | Detects motion and intrusion events                                        |
| 3   | DHT11 Temperature & Humidity Sensor | 1        | Measures environmental temperature and humidity                            |
| 4   | Breadboard                          | 1        | Prototype platform for circuit assembly                                    |
| 5   | Jumper Wires                        | 1 Set    | Electrical connections between components                                  |
| 6   | LED (Green)                         | 1        | Indicates normal system status                                             |
| 7   | LED (Red)                           | 1        | Indicates alarm or motion detection                                        |
| 8   | 220Ω Resistors                      | 2        | Current limiting for LEDs                                                  |
| 9   | USB Cable (Micro-USB/Type-C)        | 1        | Power supply and programming interface for ESP32                           |

# Results

1. The connectivity between devices was successfully verified using Ping.
2. The Telemetry Server was successfully activated with HTTP and HTTPS services enabled.
3. Security was implemented using VLAN segmentation to isolate the Control, Users, Security, and IoT networks.



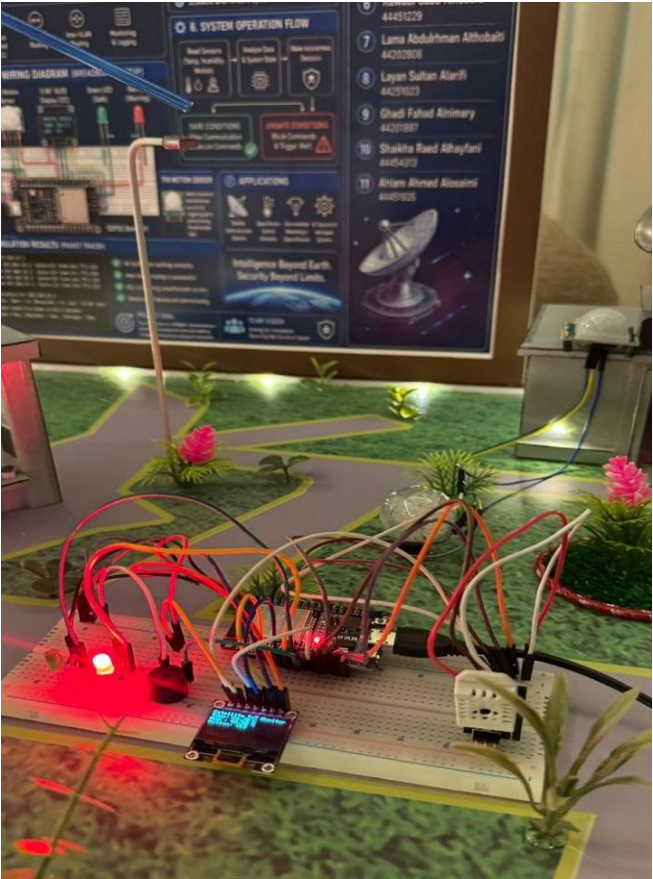
# Analysis





# Screen Shots and Comments

*All project*



*The circuit*

# Conclusion

SQAIN is a secure smart navigation system designed to protect communication and control systems in small satellites (CubeSats) and aerospace applications. The project addresses major cybersecurity challenges, including weak encryption, data interference, and limited onboard resources.

The proposed solution combines multiple security layers, such as quantum-inspired authentication, intelligent routing protocols, and lightweight security mechanisms that can operate efficiently on resource-constrained satellites.

## Expected Impact:

Prevent unauthorized access to satellite and aircraft control systems.

Improve network resilience by automatically rerouting data during cyberattacks.

Reduce security risks caused by data interference and command hijacking.

Demonstrate a low-cost, high-efficiency proof of concept (PoC) that can be expanded for real aerospace systems.

Overall, SQAIN aims to enhance the security, reliability, and resilience of future aerospace communication networks while maintaining low power and resource consumption.

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